INSTITUTO TECNOLÓGICO Y DE ESTUDIOS SUPERIORES DE MONTERREY

Campus Guadalajara



Modelación de sistemas mínimos y arquitecturas computacionales (Gpo 101)

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Sistemas mínimos

TAREA 2.

- $1263_8 \rightarrow 10$

$1 \times 8^3 + 2 \times 8^2 + 6 \times 8^1 + 3 \times 8^0 =$

$512 + 128 + 48 + 3 = 691$

- $B6A.5F_{16} \rightarrow 2$

1011 0110 1010 0101 1111$_2$

- $1011010111.001_2 \rightarrow 16$

0010 1101 0111 . 0010

$2D7.2_{16}$

- $143_{10} \rightarrow 8$

143	8
17	7
2	1
0	2

217

• $2C6.B_{16} \rightarrow 10$

$2 \times 16^2 + 12 \times 16^1 + 6 \times 16^0 . 12 \times 16^{-1} =$

$512 + 192 + 6 + 0.75 = 710.75$

• $241_8 \rightarrow 16$

$241_8 \rightarrow 2$ = 010 - 100 - 001

0|0100001 → $A1_{16}$

Convertir a decimal los valores binarios:

• 101011.01110

$2^5 + 2^3 + 2^1 + 2^0 . 2^{-2} + 2^{-3} + 2^{-4}$ → 43.4375

$32 + 8 + 2 + 1 . 4375$

• 01101.110111

$2^3 + 2^2 + 2^0 . 2^{-1} + 2^{-2} + 2^{-4} + 2^{-5} + 2^{-6}$

13.859375

• 101011110.011 $2^8 + 2^6 + 2^4 + 2^3 + 2^2 + 2^1 . 2^{-2} + 2^{-3}$

0001|0101|1110

1 5 E

350.375

"jarrar el peor

• 11011.111

$2^4+2^3+2^1+2^0 . 2^{-1}+2^{-2}+2^{-3}$

27.875

3. Convertir a binario

• 74.672

1001010.1010$_2$

74	2
37	0 ↑
18	1
9	0
4	1
2	0
1	0
0	1

0.672	2
0.344	1
0.688	0
0.376	1
0.752	0 ↓

• 118.856

1110110.1101$_2$

118	2
59	0
29	1
14	1
7	0
3	1
1	1
0	1

0.856	2
0.712	1
0.424	1
0.848	0
0.696	1

0.672	2
0.344	1
0.688	0
0.376	1
.752	0 ↓

0.856	2
0.712	1
0.424	1
0.848	0
0.696	1

• 257.875

100000001.111$_2$

257	2
128	1
64	0
32	0
16	0
8	0
4	0
2	0
1	0
0	1

0.875	2
0.75	1
0.5	1
0	1

• 2036.1147

11111110100.0001$_2$

2036	2
1018	0
509	0
254	1
127	0
69	1
31	1
15	1
7	1
3	1
1	1
0	1

0.1147	2
0.2294	0
0.4588	0
0.9176	0
0.8352	1

E191

4. 700 MB → bits

M = millon

Byte = 8 bits

$700 \times 1{,}000{,}000 \times 8 =$

5,600,000,000 bits

5. E 98, E9A, E9C, E9E, EA0, EA2, EA4 EA6

6. $2^3 \times 3 = 24 \rightarrow 2^{24} - 1 = 16,77,215$

7. word → 16 bits = 4 nibles × 3 = 12 nibles

8. 16 MB ≐ 16,000,000 Bytes

16 MB = 128,000,000 bits

q_3 bits$_s$ = 44,000 × 10 bits = 440,000 bits/s

440,000 bits × 60 s = 26,400,000 bits/min

$t_{tempo} = \frac{5 \times 10^9 \text{ bits}}{26,400,000 \text{ bits/min}} \approx 189.39 \text{ min}$

10 Ascii → hexadecimal "Sistema Mínimo"

S i s t e m a

83 105 115 116 101 109 97 32

M í n i m o

77 161 110 105 109 111

S i s t e m a

53 69 73 74 65 6D 61 20

M í n i m o

4D A1 6E 69 6D 6F

Complemento a 2

11. 127 →

127	2
63	1
31	1
15	1
7	1
3	1
1	1
0	1

01111111_c

10000000

$+ \quad 1$

10000001

¿ 100000111012 →

$2^8+2^7+2^6+2^5+2^4+2^1+2^0 =$

$= 499$

$011111001\,0$

$+ \quad 1$

$011111001\,1_c$

127 − 499 →

~~10000001~~

~~01111111~~

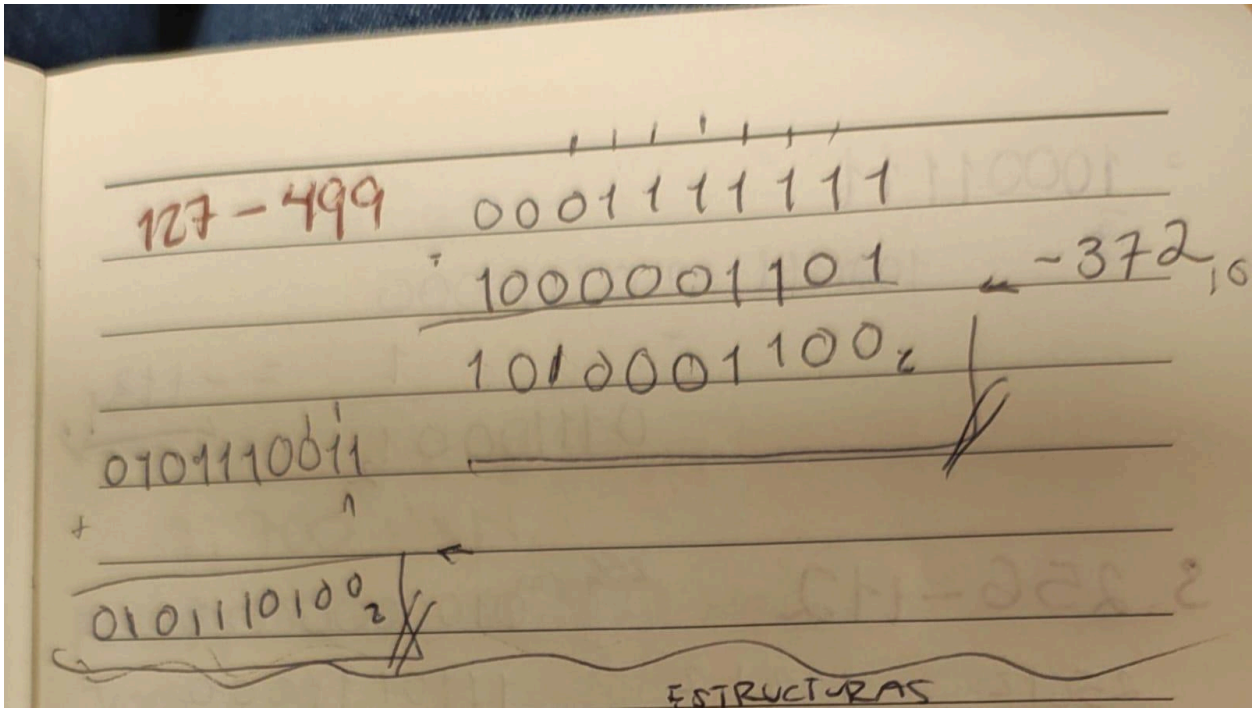
~~100000~~

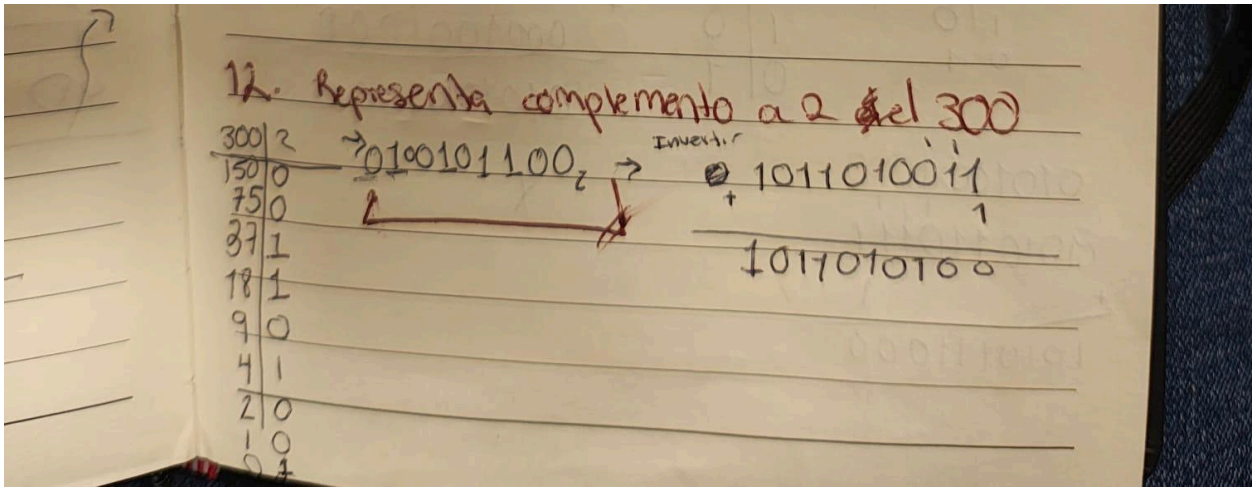
00111111111

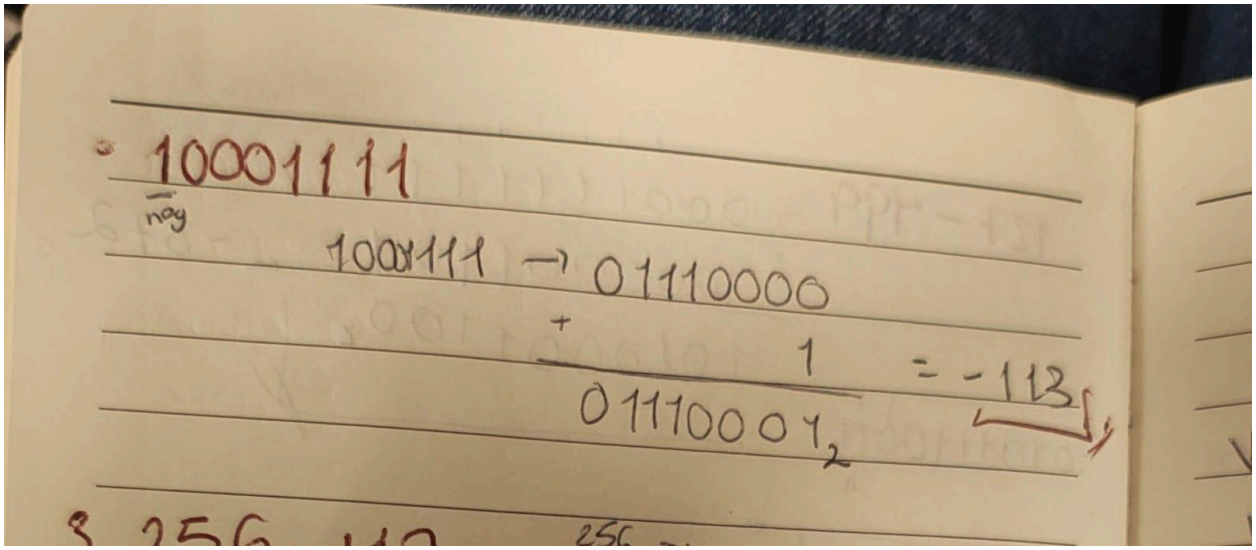
110011

$00\,11\,11\,11\,11$

$+ \quad 1\,\,11\,11\,00\,11$


127 - 499
0001111111
1000001101
-372,0
1010001100₂
0101110011
+
0101110100₂
ESTRUCTURAS


12. Representa complemento a 2 del 300
300 | 2
150 | 0
75 | 0
37 | 1
18 | 1
9 | 0
4 | 1
2 | 0
1 | 0
0 | 1
→ 0100101100₂ →
Invertir
1011010011
+ 1
1011010100


10001111
nag
1000111 → 01110000
+ 1
01110001₂
= -113

256 - 112

+256 → 0100000000

256	2
128	0
64	0
32	0
16	0
8	0
4	0
2	0
1	0
0	1

112	2
56	0
28	0
14	0
7	0
3	1
1	1
0	1

112 → 0001110000

1110001111 ↓

\+ 1 ← 1

1110010000 ←

0100000000

\+ 1110010000

0010010000

$0010010000_2 \rightarrow 144_{10}$